Emergent Partitioning in Continuous Predictive Systems:

A Variance- and Cost-Constrained Alternative to Explicit Gating

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Author Statement

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Abstract

Partitioning in predictive systems is typically implemented via explicit gating mechanisms (e.g., Mixture-of-Experts). This work demonstrates that partitioning can instead emerge from continuous, variance- and cost-constrained error minimization within a shared state-space.

Competing priors remain jointly active and are continuously weighted. Under sustained structured error, the system exhibits clustering, precision-weight collapse, and context-dependent separation, producing soft, reversible partitions without predefined boundaries.

The framework predicts smooth weight transitions, history-dependent clustering, and gradual partition formation, in contrast to the threshold discontinuities of discrete gating.

A minimal empirical protocol using multi-cue perceptual decision tasks is defined, with clear observables and falsifiability criteria.

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1. Introduction

Predictive systems must resolve competition among candidate priors. Standard approaches use explicit gating to enforce specialization via discrete selection.

These approaches impose:

- fixed partition boundaries,
- threshold-based switching,
- externally defined structure.

We instead test:

Partitioning emerges from constrained error minimization, rather than being imposed.

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2. Formal Framework

2.1 Continuous Arbitration

$$\hat{y} = \sum_i w_i P_i(x), \quad \sum_i w_i = 1$$

$$\mathcal{L} = \mathbb{E}[(y - \hat{y})^2] + \lambda C(w)$$

2.2 Governing Constraints

1. Error minimization
2. Variance sensitivity
3. Cost constraint

No gating or expert assignment is assumed.

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3. Emergent Partitioning Dynamics

Partitioning arises from:

- **Variance clustering** → coherent error regions
- **Precision collapse** → local specialization
- **Decoherence** → separation across contexts
- **Temporal persistence** → stabilization filter
- **Cost constraint** → minimal sufficient structure

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4. Continuous Arbitration

- Outputs remain **weighted composites**
- No discrete switching
- Poor priors undergo **gradual, proportional demotion**

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5. Comparison to MoE

Property	Continuous	MoE
Selection	Continuous	Discrete
Boundaries	Emergent	Fixed
Transitions	Smooth	Threshold
Adaptation	History-based	Reset-like
Structure	Learned	Imposed

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6. Empirical Protocol (v0.1)

Task

Multi-cue perceptual decision with:

- overlapping cues
- reliability control
- conflict
- context shifts

Conditions

- Baseline
- Conflict
- Reliability shift
- Context shift
- High variance

Observables

- Accuracy
- Reaction time
- Adaptation curves
- Weight inference
- Variance structure

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7. Falsifiable Predictions

Continuous model:

- smooth weight shifts

- gradual clustering
- reversible partitions

MoE:

- threshold switching
- abrupt transitions

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8. Separation Criteria

- Weight continuity
- Transition signatures
- Mixed-cue behavior
- Adaptation curves
- Variance structure
- RT / cost

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9. Early Partition Formation

Under high variance:

- initial weighting remains diffuse
- transient clusters form
- only persistent clusters stabilize
- early partitions remain reversible

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10. Discussion

Partitioning, specialization, and arbitration are unified as consequences of:

error minimization under variance and cost constraints

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11. Conclusion

Partitioning is not required, it is **discovered**.

Continuous systems naturally produce:

- adaptive specialization
- soft partitioning
- stable arbitration

without explicit gating.

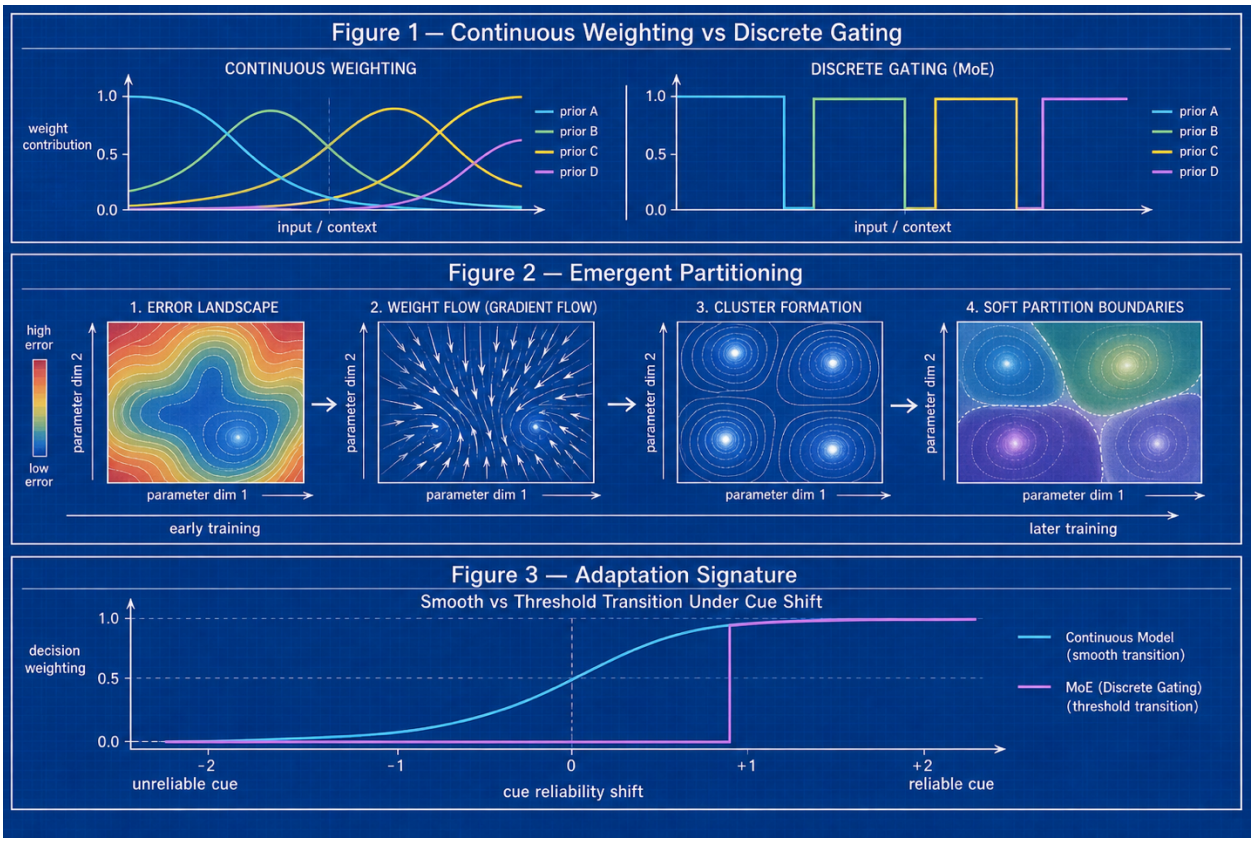

Figure 1 — Continuous Weighting vs Discrete Gating

Smooth overlapping weight curves vs step-function assignment.

Figure 2 — Emergent Partitioning

Error landscape → gradient flow → clustering → soft partitions.

Figure 3 — Adaptation Signature

Smooth vs threshold transition under cue reliability shift.

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